Numbers, Percents & Ratios

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the complex number system, where $i^2 = -1$, the expression $\frac{a+6i}{2-i}$ is equal to $4 + bi$ for some real numbers a and b. What is the value of $a + b$?

To find the value of $a + b$, first relate the two sides of the equation: $\frac{a+6i}{2-i} = 4 + bi$ Multiply both sides by $2 - i$: $a + 6i = (4 + bi)(2 - i)$ Expand the right-hand side using the FOIL method and the identity $i^2 = -1$: $(4 + bi)(2 - i) = 8 - 4i + 2bi - bi^2$ Because $i^2 = -1$, the term $-bi^2 = -b(-1) = +b$: $8 - 4i + 2bi + b = (8 + b) + (2b - 4)i$ Now equate the corresponding real and imaginary components from both sides of $a + 6i = (8 + b) + (2b - 4)i$: 1. Equating imaginary parts: $6 = 2b - 4 \Rightarrow 2b = 10 \Rightarrow b = 5$ 2. Equating real parts: $a = 8 + b \Rightarrow a = 8 + 5 = 13$ Finally, determine $a + b$: $a + b = 13 + 5 = 18$

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

On the real number line below, point A has coordinate -16 and point C has coordinate 24. Point B has coordinate k, where $-16 < k < 24$. If the ratio of the distance between A and B to the distance between B and C is $3 : 5$, what is the value of k?

First, find the total distance between points A and C: $AC = 24 - (-16) = 40$ The point B divides segment AC in the ratio $3 : 5$, meaning the segment is divided into $3 + 5 = 8$ equal parts. The length of each part is: $\frac{40}{8} = 5$ The distance from A to B corresponds to 3 parts: $AB = 3 \times 5 = 15$ Since B lies to the right of A, its coordinate k is: $k = -16 + 15 = -1$ Alternatively, set up an algebraic proportion using the coordinate distances: $\frac{k-(-16)}{24-k} = \frac{3}{5}$ $5(k + 16) = 3(24 - k)$ $5k + 80 = 72 - 3k$ $8k = -8 \Rightarrow k = -1$ Common algebraic traps and misconceptions: - 4 is the midpoint of A and C, calculated as $\frac{-16+24}{2} = 4$. - 9 results from reversing the ratio parts (using $5 : 3$ instead of $3 : 5$), giving $-16 + 25 = 9$. - 15 is the distance AB, confusing the length of the segment with the coordinate k.

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3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

On the number line shown below, point A is located at -16 and point B is located at 24. Point P lies on the line segment connecting A and B such that the ratio of the length of $\overline{AP}$ to the length of $\overline{PB}$ is $3:2$. What is the coordinate of point P?

To find the coordinate of point P, follow these steps: 1. Find the total distance between points A and B: $AB = 24 - (-16) = 24 + 16 = 40$ 2. Determine the fraction of the total distance represented by $\overline{AP}$. The ratio of the length of $\overline{AP}$ to $\overline{PB}$ is $3:2$, meaning there are $3 + 2 = 5$ equal parts in total. Thus, $\overline{AP}$ accounts for $\frac{3}{5}$ of the total distance: $AP = \frac{3}{5} \times 40 = 24$ 3. Add this distance to the starting coordinate of point A: $\text{Coordinate of } P = -16 + 24 = 8$ Check against misconceptions: - Finding the midpoint $\frac{-16+24}{2} = 4$ ignores the given $3:2$ ratio. - Reversing the ratio to $2:3$ yields $-16 + \frac{2}{5}(40) = 0$. - Calculating the length of $\overline{PB} = 16$ gives the remaining distance rather than the coordinate of P.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the distribution of the number of siblings reported by the 25 students in a homeroom class. If 1 student is selected at random from the class, what is the probability that the selected student has at least 2 siblings?

To find the probability that a randomly chosen student has at least 2 siblings: 1. Find the total number of outcomes (students): $6 + 10 + 6 + 3 = 25$ 2. Find the number of favorable outcomes. The phrase "at least 2 siblings" means having 2 or more siblings (either 2 or 3 siblings): - Number of students with 2 siblings: 6 - Number of students with 3 siblings: 3 - Favorable outcomes: $6 + 3 = 9$ 3. Calculate the probability: $P(\text{at least 2 siblings}) = \frac{9}{25}$ Common Misconceptions: - $\frac{6}{25}$ counts only students with exactly 2 siblings, failing to include those with 3. - $\frac{16}{25}$ incorrectly counts students with fewer than 2 siblings (0 or 1 sibling: $6 + 10 = 16$).

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?

To find the percentage of total hours logged by Maya: 1. Find the total number of volunteer hours logged by all four students: $\text{Total hours} = 8 + 14 + 6 + 12 = 40$ 2. Determine the fraction of hours Maya contributed: $\frac{\text{Maya's hours}}{\text{Total hours}} = \frac{14}{40}$ 3. Simplify the fraction and convert to a percentage: $\frac{14}{40} = \frac{7}{20} = \frac{35}{100} = 35\%$

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the number of books read by 4 students—Student A, Student B, Student C, and Student D—during summer break. Student A read 3 books, Student B read 5 books, Student C read 8 books, and Student D read 12 books. What is the average (arithmetic mean) number of books read by the 4 students?

To find the average (arithmetic mean) number of books read by the 4 students, find the sum of all books read and divide by the total number of students (4): $\text{Mean} = \frac{3+5+8+12}{4} = \frac{28}{4} = 7.0$

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?

We can solve for k using two algebraic methods: Method 1: Symmetry and Factored Form The axis of symmetry of the parabola passes vertically through the vertex, so its equation is $x = 1$. Since $(4, 0)$ is an x-intercept, its horizontal distance from the axis of symmetry is $4 - 1 = 3$. By symmetry, the other x-intercept must lie 3 units to the left of $x = 1$, which is at $x = 1 - 3 = -2$. Since the leading coefficient of $f(x) = -x^2 + bx + c$ is $a = -1$, we can write $f(x)$ in factored form:
$f(x) = -(x - 4)(x - (-2)) = -(x - 4)(x + 2)$ The value of k is the y-coordinate at the vertex $x = 1$:
$k = f(1) = -(1 - 4)(1 + 2) = -(-3)(3) = 9$ Method 2: Vertex Formula and Standard Form For $f(x) = -x^2 + bx + c$, the x-coordinate of the vertex is given by: $x = -\frac{b}{2a} = -\frac{b}{2(-1)} = \frac{b}{2}$ Setting this equal to 1 yields $b = 2$. Since $(4, 0)$ lies on the parabola, substitute $x = 4$ and $y = 0$:
$-(4)^2 + 2(4) + c = 0 \Rightarrow -16 + 8 + c = 0 \Rightarrow c = 8$ Thus, the function is $f(x) = -x^2 + 2x + 8$. Finally, find $k = f(1)$: $k = -(1)^2 + 2(1) + 8 = -1 + 2 + 8 = 9$ Common Traps: - Confusing the y-intercept with the vertex gives $c = 8$. - Forgetting that the leading coefficient is -1 can cause sign errors during expansion.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(3, 2)$ and passing through the point $(1, 6)$. What is the value of b?

1. Write the quadratic function in vertex form: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. 2. Given the vertex $(3, 2)$, the equation becomes: $f(x) = a(x - 3)^2 + 2$ 3. Use the given point $(1, 6)$ to solve for the parameter a: $6 = a(1 - 3)^2 + 2$ $6 = a(-2)^2 + 2$ $6 = 4a + 2 \Rightarrow 4a = 4 \Rightarrow a = 1$ 4. Substitute $a = 1$ into the vertex form and expand to standard form $f(x) = ax^2 + bx + c$: $f(x) = 1(x - 3)^2 + 2 = (x^2 - 6x + 9) + 2 = x^2 - 6x + 11$ 5. Comparing coefficients with $ax^2 + bx + c$, we identify $a = 1$, $b = -6$, and $c = 11$. Therefore, $b = -6$. Common traps: - Incorrectly assuming $b = -h = -3$ or $b = h = 3$ directly from the vertex coordinates without expanding into standard form. - Making a sign error when expanding $(x - 3)^2$ as $x^2 + 6x + 9$, which leads to $b = 6$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the right triangle ΔABC shown below, the right angle is at vertex C. The lengths of the sides are given in inches: $AC = 12$, $BC = 5$, and $AB = 13$. What is the value of $\cos B$?

In any right triangle, the cosine of an acute angle is defined as the ratio of the length of the side adjacent to the angle to the length of the hypotenuse: $\cos B = \frac{\text{length of side adjacent to } B}{\text{length of hypotenuse}} = \frac{BC}{AB}$ From the given triangle: - The side adjacent to $\angle B$ is $BC = 5$. - The hypotenuse is $AB = 13$. Therefore, $\cos B = \frac{5}{13}$. Common Misconceptions: - $\frac{12}{13}$ represents $\sin B = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{AC}{AB}$ or $\cos A$. - $\frac{5}{12}$ represents $\frac{\text{adjacent}}{\text{opposite}} = \cot B$ (or $\tan A$). - $\frac{12}{5}$ represents $\tan B = \frac{\text{opposite}}{\text{adjacent}} = \frac{AC}{BC}$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC shown below, the measure of $\angle A$ is 65° and the measure of $\angle B$ is 45°. What is the value of x ?

According to the Triangle Angle Sum Theorem, the sum of the interior angle measures of any triangle is 180°. For ΔABC: $m\angle A + m\angle B + m\angle C = 180^\circ$ Substitute the given values: $65^\circ + 45^\circ + x^\circ = 180^\circ$ $110^\circ + x^\circ = 180^\circ$ $x = 180 - 110 = 70$ Common Misconceptions: - 110 results from merely calculating the sum $65 + 45$ without subtracting from 180°. - 65 results from incorrectly assuming the triangle is isosceles with $\angle C \cong \angle A$. - 115 results from subtracting only a single angle $(180 - 65)$ rather than the sum of both given angles.

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11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of the quadratic function $f(x) = a(x-h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c?

1. Identify the vertex form of the parabola: Since the vertex is $(h, k) = (2, -4)$, the function is: $f(x) = a(x-2)^2 - 4$ 2. Solve for the leading parameter a using the given y-intercept $(0, -2)$: $-2 = a(0-2)^2 - 4$ $-2 = 4a - 4$ $2 = 4a \Rightarrow a = \frac{1}{2}$ 3. Determine the full equation of the function: $f(x) = \frac{1}{2}(x-2)^2 - 4$ 4. Find c by substituting $x = 6$: $c = f(6) = \frac{1}{2}(6-2)^2 - 4 = \frac{1}{2}(16) - 4 = 8 - 4 = 4$ Misconception traps: - Assuming $a = 1$ leads to $(6-2)^2 - 4 = 12$ (Choice D). - Forgetting to subtract $k = 4$ leads to $\frac{1}{2}(16) = 8$ (Choice C).

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A right triangle has legs 9 and 12. Find the hypotenuse.

Apply the Pythagorean Theorem $a^2 + b^2 = c^2$: $c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = \mathbf{15}$ ACT Scaling Shortcut: Notice that both legs are multiples of 3: $9 = 3 \times 3$ and $12 = 3 \times 4$. This is a $3-4-5$ Pythagorean triple scaled by factor 3: $3 \times 5 = \mathbf{15}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A triangle has angles 50° and 60°. Find the third.

By the Triangle Angle Sum Theorem, the interior angles of any planar triangle sum to 180°: $\theta_1 + \theta_2 + \theta_3 = 180^\circ$ $50^\circ + 60^\circ + \theta_3 = 180^\circ \Rightarrow 110^\circ + \theta_3 = 180^\circ$ $\theta_3 = 180^\circ - 110^\circ = \mathbf{70}^\circ$

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the area of a circle with radius 5.

Apply the circle area formula $A = \pi r^2$: $A = \pi(5)^2 = \mathbf{25\pi} \approx 78.54$ Trap Warning: Do NOT confuse area (πr^2) with circumference ($2\pi r = 10\pi$). Most ACT answer choices leave π exact.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

To have both zeros of the upward-opening parabola $f(x) = x^2 - 2kx + (3k + 10)$ be real, distinct, and strictly greater than 2, three conditions must hold simultaneously: 1. Positive Discriminant ($\Delta > 0$): $\Delta = (-2k)^2 - 4(1)(3k + 10) = 4k^2 - 12k - 40 = 4(k - 5)(k + 2) > 0$. This requires $k > 5$ or $k < -2$. 2. Vertex Position ($x_v > 2$): The axis of symmetry of $f(x)$ is at $x = -\frac{-2k}{2(1)} = k$. For both zeros to lie to the right of 2, the midpoint of the zeros must satisfy: $k > 2$. Intersecting $k > 2$ with $k > 5$ or $k < -2$ narrows the condition to $k > 5$. 3. Boundary Value ($f(2) > 0$): Since the parabola opens upward, having 2 to the left of both zeros requires: $f(2) = 2^2 - 2k(2) + 3k + 10 = 14 - k > 0 \Rightarrow k < 14$. Combining all inequalities yields: $5 < k < 14$. The possible integer values for k are $\{6, 7, 8, 9, 10, 11, 12, 13\}$. Counting these integers gives $13 - 6 + 1 = 8$. *Common Misconceptions:* - Forgetting the discriminant restriction $\Delta > 0$ leads to $2 < k < 14$, which falsely gives 11 integers. - Including $k = 5$ (where the zeros are not distinct) or $k = 14$ (where one zero is exactly 2) leads to 9 integers. - An off-by-one counting error may produce 7.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

The graph of the quadratic function $f(x) = 2x^2 - 12x + c$ in the standard (x, y) coordinate plane passes through the point $(1, -3)$. What is the minimum value of $f(x)$?

Step 1: Determine the unknown parameter c by substituting the coordinates of the given point $(1, -3)$ into the function equation: $f(1) = 2(1)^2 - 12(1) + c = -3$ $2 - 12 + c = -3$ $-10 + c = -3 \Rightarrow c = 7$ Step 2: Write the complete function: $f(x) = 2x^2 - 12x + 7$ Step 3: Find the x-coordinate of the vertex. For a parabola $y = ax^2 + bx + c$, the vertex occurs at $x = -\frac{b}{2a}$: $x = -\frac{-12}{2(2)} = \frac{12}{4} = 3$ Step 4: Find the minimum value by evaluating $f(3)$: $f(3) = 2(3)^2 - 12(3) + 7 = 2(9) - 36 + 7 = 18 - 36 + 7 = -11$ Common traps: - Selecting 3 represents the x-coordinate of the vertex rather than the minimum value (the y-coordinate). - Selecting 7 represents the value of the parameter c (the y-intercept). - Selecting -3 is simply repeating the y-coordinate of the given point $(1, -3)$.

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

In the figure below, ΔABC is a right triangle with right angle at C, and altitude $\overline{CD}$ of length h is perpendicular to hypotenuse $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the area of ΔABC?

To find the area of ΔABC, we need the base AB and the corresponding altitude $CD = h$. 1. Find the altitude h: By the Geometric Mean (Altitude) Theorem for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $h^2 = AD \cdot DB = 4 \cdot 9 = 36$ Taking the square root gives: $h = \sqrt{36} = 6$ 2. Find the total length of base $\overline{AB}$: $AB = AD + DB = 4 + 9 = 13$ 3. Calculate the area of ΔABC: $\text{Area} = \frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot 13 \cdot 6 = 39$ Common traps: - Finding the area of only sub-triangle ΔCDB: $\frac{1}{2} \cdot 9 \cdot 6 = 27$. - Confusing the altitude squared with the area: $4 \cdot 9 = 36$. - Forgetting the factor of $\frac{1}{2}$ in the area formula: $13 \cdot 6 = 78$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

If $\cos\theta = \frac{12}{13}$ and $0^\circ < \theta < 90^\circ$, what is the value of $\tan\theta$?

In a right triangle with acute angle θ, the cosine ratio is defined as: $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{12}{13}$ We can represent the adjacent leg as 12 and the hypotenuse as 13. Using the Pythagorean theorem to find the length of the opposite leg: $\text{opposite} = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5$ The tangent ratio is defined as: $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$ - $\frac{5}{13}$ is incorrect because it represents $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$. - $\frac{12}{5}$ is incorrect because it is the reciprocal ratio $\cot\theta = \frac{\text{adjacent}}{\text{opposite}}$. - $\frac{13}{12}$ is incorrect because it represents $\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

In the figure below, ΔABC is a right triangle with the right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$. The length of $\overline{AD}$ is 4 units and the length of $\overline{DC}$ is 9 units. What is the area, in square units, of ΔABC?

To find the area of ΔABC, we need the base AC and the altitude BD. 1. Find the length of altitude $\overline{BD}$: By the Right-Triangle Altitude Theorem (Geometric Mean Theorem), the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $BD^2 = AD \cdot DC$
$BD^2 = 4 \cdot 9 = 36 \Rightarrow BD = \sqrt{36} = 6$ 2. Find the length of hypotenuse $\overline{AC}$: $AC = AD + DC = 4 + 9 = 13$ 3. Calculate the area of ΔABC: $\text{Area} = \frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot AC \cdot BD = \frac{1}{2} \cdot 13 \cdot 6 = 39$ Alternatively, sum the areas of the two smaller right triangles: $\text{Area}(\Delta ABD) = \frac{1}{2}(4)(6) = 12$ $\text{Area}(\Delta BCD) = \frac{1}{2}(9)(6) = 27$ $\text{Total Area} = 12 + 27 = 39$

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20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

In right triangle ΔABC, the measure of $\angle C$ is 90°. The length of side AC is 12 and the length of side BC is 5. What is the value of $\cos A$?

In right triangle ΔABC with the right angle at C, side AC is the leg adjacent to $\angle A$, side BC is the leg opposite to $\angle A$, and side AB is the hypotenuse. First, apply the Pythagorean theorem to find the length of the hypotenuse AB: $AB = \sqrt{AC^2 + BC^2} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ By the definition of cosine in a right triangle: $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AC}{AB} = \frac{12}{13}$ - Choosing $\frac{5}{13}$ confuses cosine with sine ($\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$). - Choosing $\frac{5}{12}$ confuses cosine with tangent ($\tan A = \frac{\text{opposite}}{\text{adjacent}}$). - Choosing $\frac{12}{5}$ calculates cotangent ($\cot A = \frac{\text{adjacent}}{\text{opposite}}$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of the numerical expression $|-14| - 3|5-9| + (-2)^3$?

a. -6 ✓

b. -10

c. 2

d. 14

$|-14| = 14$, $|5-9| = |-4| = 4$, and $(-2)^3 = -8$. Substituting gives $14 - 3(4) + (-8) = 14 - 12 - 8 = -6$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If p is a positive prime number and q is a negative integer, which of the following values must always be negative?

a. $p - q$

b. $q - p$ ✓

c. $p^2 + q^2$

d. $\frac{p}{q^2}$

Since $p > 0$ and $q < 0$, $-p < 0$. Therefore, $q - p$ is the sum of two negative numbers, which is always negative.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the greatest common factor of 72, 108, and 180?

a. 18

b. 24

c. 36 ✓

d. 72

Prime factorizations: $72 = 2^3 \times 3^2$, $108 = 2^2 \times 3^3$, $180 = 2^2 \times 3^2 \times 5$. GCF $= 2^2 \times 3^2 = 4 \times 9 = 36$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On a blueprint, a scale of $\frac{1}{4}$ inch represents 3 feet. If a conference room has a length of 36 feet, what is its length on the blueprint?

a. 2.5 inches

b. 3 **inches** ✓

c. 4.5 inches

d. 6 inches

Set up the proportion: $\frac{0.25 \text{ in}}{3 \text{ ft}} = \frac{x \text{ in}}{36 \text{ ft}}$. Cross-multiplying yields $3x = 0.25 \times 36 = 9 \Rightarrow x = 3$ inches.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A car travels 180 miles on 6 gallons of fuel. At the same rate of fuel consumption, how many gallons of fuel are required to travel 450 miles?

a. 12

b. 15 ✓

c. 16.5

d. 18

Fuel efficiency is $\frac{180}{6} = 30$ miles per gallon. For 450 miles: $\frac{450}{30} = 15$ gallons.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The ratio of copper to zinc in a brass alloy is $5 : 3$ by weight. If a piece of brass weighs 56 kilograms, how many kilograms of copper does it contain?

a. 21

b. 30

c. 35 ✓

d. 40

Total ratio parts $= 5 + 3 = 8$. Each part weighs $\frac{56}{8} = 7$ kg. Weight of copper $= 5 \times 7 = 35$ kg.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A store purchases a bicycle for 150 and marks up the price by 40%. During a weekend clearance sale, the bicycle is discounted by 20%. What is the final sale price?

a. 168 ✓

b. 174

c. 180

d. 192

Retail price after 40% markup $= 150 \times 1.40 = 210$. After 20% discount, final price $= 210 \times 0.80 = 168$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If 35% of a number N is equal to 84, what is 60% of N?

a. 120

b. 144 ✓

c. 168

d. 240

$0.35N = 84 \Rightarrow N = \frac{84}{0.35} = 240$. Then 60% of $240 = 0.60 \times 240 = 144$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The price of an airline ticket increased from 250 to 320. What was the percent increase in the ticket price?

a. 22%

b. 25%

c. 28% ✓

d. 30%

Increase $= 320 - 250 = 70$. Percent increase $= \frac{70}{250} \times 100\% = \frac{7}{25} \times 100\% = 28\%$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of $(3.2 \times 10^{7}) \times (2.5 \times 10^{-3})$ expressed in standard scientific notation?

a. 8.0×10^{4} ✓

b. 8.0×10^{5}

c. 8.0×10^{3}

d. 0.8×10^{5}

Multiply coefficients: $3.2 \times 2.5 = 8.0$. Add exponents: $10^{7+(-3)} = 10^{4}$. The result is 8.0×10^{4}.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following is equivalent to $\frac{x^5 \cdot x^{-2}}{(x^3)^2}$ for all positive values of x?

a. x^{-3} ✓

b. x^{-1}

c. x^0

d. x^2

Numerator $= x^{5-2} = x^3$. Denominator $= x^{3\times 2} = x^6$. Quotient $= \frac{x^3}{x^6} = x^{3-6} = x^{-3}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the radical expression $\sqrt{75} - 2\sqrt{27} + \sqrt{48}$.

a. $2\sqrt{3}$

b. $3\sqrt{3}$ ✓

c. $4\sqrt{3}$

d. $5\sqrt{3}$

Simplify each radical: $\sqrt{75} = 5\sqrt{3}$, $2\sqrt{27} = 2(3\sqrt{3}) = 6\sqrt{3}$, and $\sqrt{48} = 4\sqrt{3}$. Combining yields $(5 - 6 + 4)\sqrt{3} = 3\sqrt{3}$.

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33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Every student in the graduating class of Oakridge High School plans to do exactly one of the following after graduation: attend a 2-year college, attend a 4-year university, or enter the workforce directly. • The ratio of the number of students planning to attend a 2-year college to the number of students planning to attend a 4-year university is $3:5$. • Of the students planning to attend a 2-year college, exactly 80% will attend an in-state college. • Of the students planning to attend a 4-year university, exactly 72% will attend an in-state university. • The total number of students planning to attend an in-state college or university accounts for exactly 45% of the entire graduating class. If the number of students in every category and subcategory must be a positive integer, what is the minimum possible number of students in the graduating class?

a. 40

b. 100

c. 120

d. **200 ✓**

Let N be the total number of students in the graduating class. 1. **Express college counts in terms of a multiplier x:** The ratio of students attending a 2-year college to those attending a 4-year university is $3:5$. Thus, for some positive value x: $C_2 = 3x \quad \text{and} \quad C_4 = 5x$ Since both $3x$ and $5x$ must be integers, and $\gcd(3,5) = 1$, $x = 2(5x) - 3(3x)$ must be a positive integer. 2. **Determine the number of in-state students:** - In-state 2-year college students: $0.80(3x) = 2.4x = \frac{12}{5}x$. - In-state 4-year university students: $0.72(5x) = 3.6x = \frac{18}{5}x$. For both quantities to be integers, 5 must divide x, meaning x must be a multiple of 5. 3. **Relate in-state students to the total class size N:** The total number of in-state students is: $2.4x + 3.6x = 6x$ We are given that this total is 45% of N: $6x = 0.45N = \frac{9}{20}N \Rightarrow N = \frac{20}{9}(6x) = \frac{40}{3}x$ 4. **Apply integer constraints:** For $N = \frac{40}{3}x$ to be an integer, 3 must divide x (since $\gcd(40,3) = 1$). Since x must be divisible by both 5 and 3, and $\gcd(5,3) = 1$, x must be a multiple of: $\text{lcm}(3,5) = 15$ 5. **Find the minimum class size:** The smallest positive integer value for x is $x = 15$. Substituting $x = 15$ into the expression for N: $N = \frac{40}{3}(15) = 40 \times 5 = 200$ **Check of subcategories for $x = 15$:** - 2-year college: 45 (36 in-state, 9 out-of-state) - 4-year university: 75 (54 in-state, 21 out-of-state) - In-state total: $36 + 54 = 90$, which is $\frac{90}{200} = 45\%$ - Workforce: $200 - (45 + 75) = 80$ All counts are positive integers. The minimum possible number of students is 200. **Common Misconceptions:** - Choice A (40): Results from setting $x = 3$ to cancel the denominator of 3, ignoring the requirement that x must also be a multiple of 5 (which would yield 7.2 in-state 2-year students). - Choice B (100): Results from assuming a baseline of 100 students simply because percentages are used, which yields fractional student counts (e.g., $x = 7.5$ and 22.5 students in 2-year colleges). - Choice C (120): Mistaking the total number of college-bound students ($45 + 75 = 120$) for the entire graduating class.

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following is a prime number?

a. 43 ✓

b. 36

c. 38

d. 44

43 has no positive divisors other than 1 and itself, so it is prime; the other choices each factor into smaller integers.

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A boutique owner purchases a designer jacket at a wholesale price of W dollars and marks up the price by $p\%$. During an end-of-season clearance sale, this marked-up price is discounted by $p\%$. The final clearance price is $x84lessthantheoriginalwholesaleprice, andtheclearancepriceis\336$ less than the marked-up price. What is the value of W, the original wholesale price?

a. \672

b. \756 ✓

c. \1,008

d. \1,344

Let $r = \frac{p}{100}$. The marked-up price, M, is given by: $M = W(1 + r)$ The clearance sale price, S, is discounted by $p\%$ from M: $S = M(1 - r) = W(1 + r)(1 - r) = W(1 - r^2)$ We are given two pieces of information: 1. The clearance price is $x84lessthanthewholesaleprice$: W - S = 84 \implies W - W(1 - r^2) = 84 \implies W r^2 = 842. $Theclearancepriceis\336$ less than the marked-up price: $M - S = 336 \Rightarrow W(1 + r) - W(1 - r^2) = 336$ Expanding the left-hand side yields: $Wr + Wr^2 = 336$ Substitute $Wr^2 = 84$ into this equation: $Wr + 84 = 336 \Rightarrow Wr = 252$ Now divide Wr^2 by Wr to solve for r: $\frac{Wr^2}{Wr} = \frac{84}{252} \Rightarrow r = \frac{1}{3}$ Finally, solve for W: $W \cdot \frac{1}{3} = 252 \Rightarrow W = 252 \times 3 = 756$ Misconception checks: - Choice A $(\672)istheclearancesaleprice$S = 756(1 - 1/9) = 672$x1,008)$ is the marked-up price $M = 756(1 + 1/3) = 1,008$. - Choice D $(\1,344)arisesifoneincorrectlysets$W r = 336, $leadingto$r = \frac{84}{336} = \frac{1}{4}andW = 336 \times 4 = 1,344.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a high school choir of 60 students, 40% are sopranos, 25% are altos, and the remaining students are tenors or basses. How many students in the choir are tenors or basses?

a. 15

b. **21** ✓

c. 24

d. 39

First, find the percentage of students in the choir who are tenors or basses by subtracting the percentages of sopranos and altos from 100%: $100\% - (40\% + 25\%) = 100\% - 65\% = 35\%$ Next, calculate 35% of the total 60 choir students: $0.35 \times 60 = 21$ Therefore, there are 21 students who are tenors or basses. Common misconceptions: - 15 corresponds only to the number of altos (0.25×60). - 24 corresponds only to the number of sopranos (0.40×60). - 39 corresponds to the combined number of sopranos and altos (0.65×60).

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A high school track team has 45 members, and the ratio of sprinters to distance runners is 4 to 5. If every member is either a sprinter or a distance runner, how many members of the team are distance runners?

a. 20

b. 25 ✓

c. 30

d. 36

The ratio of sprinters to distance runners is $4:5$, meaning the team is divided into $4 + 5 = 9$ equal parts. To find the size of each part: $\frac{45}{9} = 5$ Since distance runners represent 5 of these equal parts, the number of distance runners is: $5 \times 5 = 25$ Common errors: - 20 represents the number of sprinters (4×5). - 36 incorrectly takes $\frac{4}{5} \times 45$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

A car travels at 7 miles per hour for 4 hours. How far does it travel?

a. 66 mi

b. 39 mi

c. **28 mi** ✓

d. 18 mi

Applying Numbers, Percents & Ratios principles, distance $= 7 \times 4 = 28$ miles. Therefore, the verified correct option is 28 mi.

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

A car travels at 6 miles per hour for 2 hours. How far does it travel?

a. 1 mi

b. 60 mi

c. **12 mi ✓**

d. 23 mi

Applying Numbers, Percents & Ratios principles, distance $= 6 \times 2 = 12$ miles. Therefore, the verified correct option is 12 mi.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

If 3 apples cost $x6$, what is the cost per apple?

a. $x32$

b. $x13$

c. **$x2$ ✓**

d. $x21$

Applying Numbers, Percents & Ratios principles, $6 \div 3 = 2$ dollars per apple. Therefore, the verified correct option is $x2$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

If 11 boxes cost $x99$, what is the cost per box?

a. $x31$

b. $x21$

c. **$x9$ ✓**

d. $x38$

Applying Numbers, Percents & Ratios principles, $99 \div 11 = 9$ dollars per box. Therefore, the verified correct option is $x9$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A car travels at 2 miles per hour for 4 hours. How far does it travel?

a. 24 mi

b. **8 mi ✓**

c. 16 mi

d. 9 mi

Applying Numbers, Percents & Ratios principles, distance $= 2 \times 4 = 8$ miles. Therefore, the verified correct option is 8 mi.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

Evaluate: $\frac{4}{6} + \frac{1}{3} \times \frac{1}{6}$.

a. $\frac{17}{50}$

b. $\frac{1}{6}$

c. $\frac{13}{18}$ ✓

d. $\frac{1}{2}$

$\frac{1}{3} \times \frac{1}{6} = \frac{1}{18}$, then $\frac{4}{6} + \frac{1}{18} = \frac{13}{18}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Evaluate: $\frac{1}{4} + \frac{1}{5} \times \frac{1}{4}$.

a. $\frac{7}{12}$

b. $\frac{3}{4}$

c. $\frac{3}{25}$

d. $\frac{3}{10}$ ✓

$\frac{1}{5} \times \frac{1}{4} = \frac{1}{20}$, then $\frac{1}{4} + \frac{1}{20} = \frac{3}{10}$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

Evaluate: $\frac{2}{6} + \frac{1}{2} \times \frac{4}{7}$.

a. 1

b. $\frac{22}{25}$

c. $\frac{6}{25}$

d. $\frac{13}{21}$ ✓

$\frac{1}{2} \times \frac{4}{7} = \frac{2}{7}$, then $\frac{2}{6} + \frac{2}{7} = \frac{13}{21}$.

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